

Day 59 - Intro to Network Automation

Wednesday, September 9, 2026

8:37 PM

Network Automation Info

- In the traditional model, engineers manage devices one at a time by connecting to their CLI via SSH.
 - This has a few downsides to it, including:
 - Typos and other small mistakes are common.
 - It is time-consuming and very inefficient in large-scale networks.
 - It is difficult to ensure that all devices adhere to the organization's standard configurations.
- Meanwhile, on the side of the automated model, tools (ie. SDN, **Ansible**, **Python scripts**, etc.) are used to setup most of the configurations on network devices.
 - This includes a few benefits, such as:
 - Human error (typos etc.) is reduced.
 - Networks become much more scalable.
 - New deployments, network-wide changes, and troubleshooting can be implemented in a fraction of the time.
 - Network-wide policy compliance can be assured (standard configurations, software versions, etc)
 - The improved efficiency of network operations reduces the opex (operating expenses) of the network. Each task requires fewer man-hours.

Logical Planes

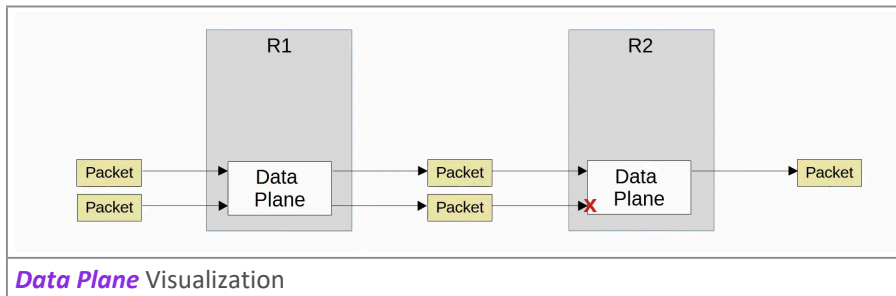
The various functions of network devices can be logically divided up into **planes**:

1. **Data plane**
2. **Control plane**
3. **Management plane**

1. Data Plane

All tasks involved in forwarding user data/traffic from one interface to another are part of the **Data plane**.

- A router receives a message, looks for the most specific matching route in its routing table, and forwards it out of the appropriate interface to the next hop.
- It also de-encapsulates the original **Layer 2** header, and re-encapsulates with a new header destined for the **next-hop's** MAC address.
- A switch receives a message, looks at the **destination** MAC address, and forwards it out of the appropriate interface (or floods it). This includes functions like adding or removing **802.1Q** VLAN tags.
- **NAT** (changing the **source/destination** addresses before forwarding) is part of the **Data plane**.
- Deciding to forward or discard messages due to **ACLs**, **port security**, etc. is part of the data plane.



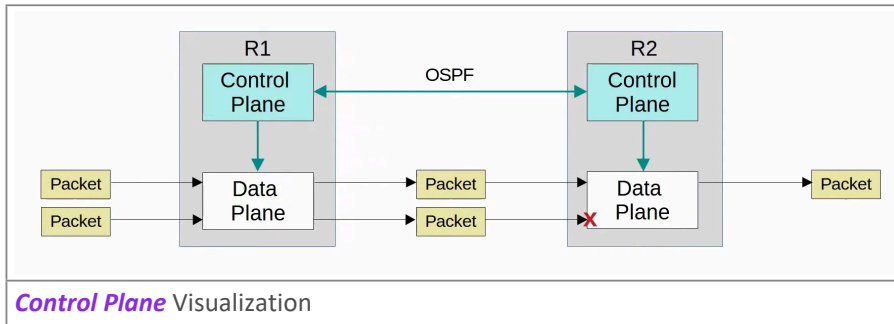
2. Control Plane

The **Control plane** contains functions that decide forwarding decisions for the **Data plane**.

- A device makes its forwarding decisions through tables, like **routing tables**, **MAC address tables**, **ARP tables**, etc.
- The **Control plane** basically **controls** what the **data plane** does, for example by building the router's routing table.
- It performs overhead work.
 - **OSPF** itself doesn't forward data packets, but it informs the **data plane** on how packets should be

forwarded.

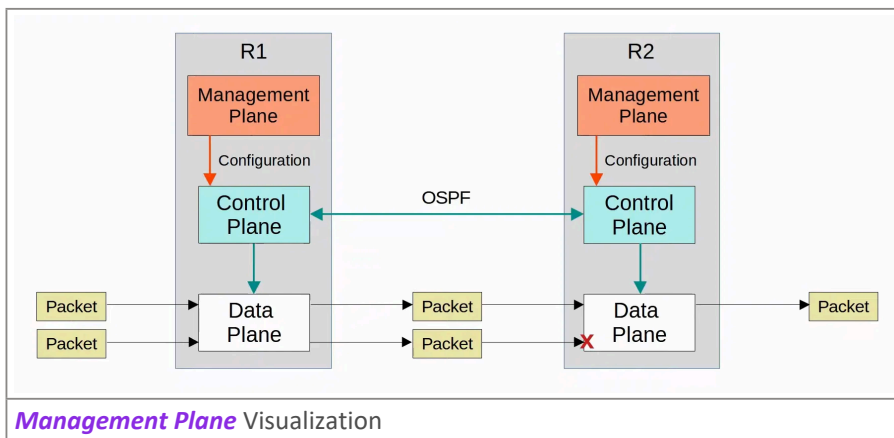
- **STP** itself isn't directly involved in the process of forwarding frames, but it informs the **Data plane** about which interfaces should and shouldn't be used to forward frames.
- **ARP** messages aren't user data, but they are used to build an **ARP** table which is used in the process of forwarding.



3. Management Plane

Similar to the **Control plane**, the **Management plane** performs overhead work. It also doesn't directly affect the forwarding of messages in the **data plane**.

- The **Management plane** consists of protocols that are used to manage devices.
 - **SSH/Telnet**, used to connect to the CLI of a device to manage/configure it.
 - **Syslog**, used to keep logs of events that occur on the device.
 - **SNMP**, used to monitor the operations of the device.
 - **NTP**, used to maintain accurate time on the device.



Traditional Networking Layout

- In traditional networking, the **Data plane** and **Control plane** are both distributed. Each device has its own **Data plane** and its own **Control plane**. The planes are 'distributed' throughout the network.

More information on the planes

- The **Data plane** is the reason why we buy routers and switches, to forward messages.
- However, the **Data plane** cannot function by itself and that's why both the **Control plane** and the **Management plane** are necessary to enable it to do its job.

Logical Planes - Special Hardware

The operations of the **Management plane** and **Control plane** are usually managed by the CPU.

- However, the CPU is not desirable for **Data plane** operations because CPU processing is slow (relatively speaking).
- That's why a specialized hardware **Application-Specific Integrated Circuit (ASIC)** is used. **ASICs** are chips built for specific purposes.
- **ASICs** are used in Switches for example, they do the following:
 - When a frame is received, the **ASIC** is responsible for the switching logic.
 - The MAC address table is stored in a kind of memory called **Ternary Content-Addressable Memory (TCAM)**.
 - MAC address tables are also commonly called **CAM tables**.
 - The **ASIC** feeds the **destination** MAC address of the frame into the **TCAM**, which returns the matching MAC

- address table entry.
 - The frame is then forwarded out of the appropriate interface.
- Modern routers also use a similar hardware **Data plane**: An **ASIC** designed for forwarding logic, and tables stored in **TCAM**.

Summary of the above:

- When a device receives **Control/Management** traffic (destined for itself), it'll be processed in the CPU.
- When a device receives **Data** traffic, which should pass through the device, it is processed by the **ASIC** for maximum speed.

Software-Defined Networking (SDN)

(You can find more info about SDN in [Day 62 - Software-Defined Networking](#))

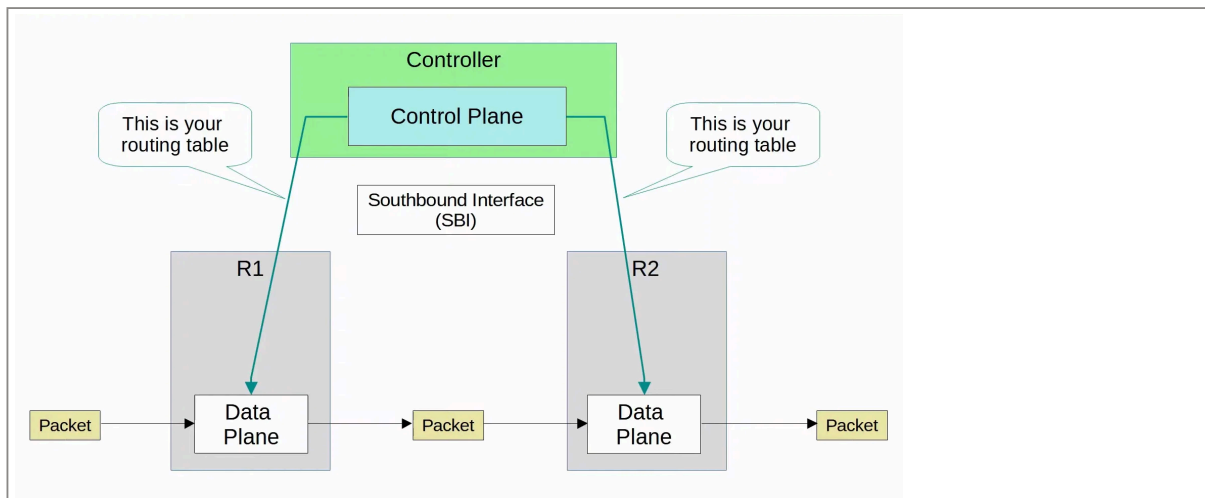
Software-Defined Networking (SDN) is an approach to networking that centralizes the **Control plane** into an application called a Controller

- This is similar to **Wireless LAN Controllers (WLCs)**.
- SDN** is also called **Software-Defined Architecture (SDA)** or **Controller-Based Networking**.
- Traditional **Control planes** use a distributed architecture.
 - For example, each router in the network runs **OSPF** and the routers share routing information and then calculate their preferred routes to each **destination**.
- An **SDN Controller** centralizes **Control plane** functions (for example, calculating routes).
- The **Controller** can interact programmatically with the network devices using **Application Programming Interfaces (APIs)**.

Southbound Interface (SBI)

The **Southbound Interface (SBI)** is used for communications between the Controller and the network devices it controls.

- It typically consists of a communication protocol and an **API**.
- APIs** facilitate data exchanges between programs.
 - Data is exchanged between the **Controller** and the network devices.
 - An **API** on the network devices allows the **Controller** to access information on the devices, control their **data plane** tables, etc.
- Using the **SBI**, the **Controller** communicates with the managed devices and gathers information about them, such as:
 - The devices in the network
 - The topology of the network (how devices are connected together)
 - The available interfaces on each device
 - Their configurations
- SBI** examples:
 - OpenFlow
 - Cisco OpFlex
 - Cisco onePK (Open Network Environment Platform Kit)
 - Netconf

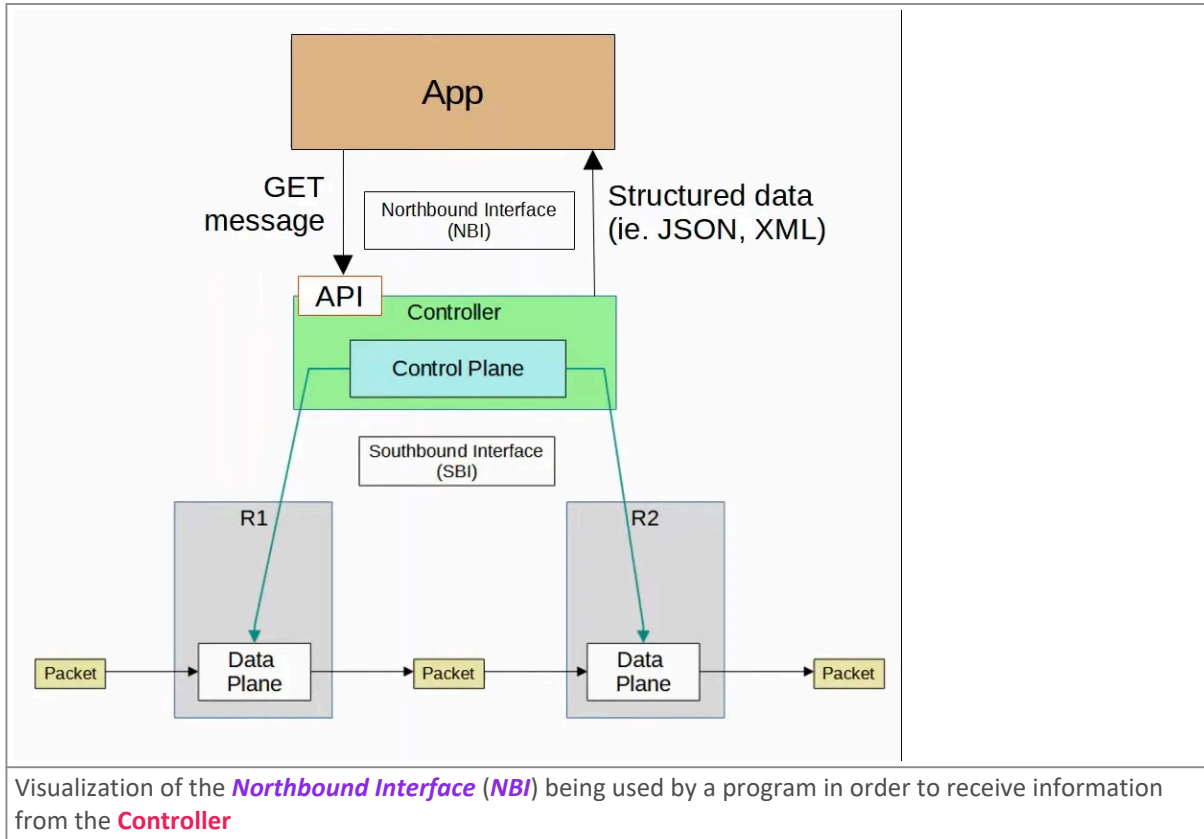


Visualization of the **Control plane** being centralized through the **SDN Controller** (Communication with network devices is done through the **Southbound Interface (SBI)**)

Northbound Interface (NBI)

The **Northbound Interface (NBI)** is what allows us to interact with the Controller, access the data it gathers, program it, and make changes to the network via the **SBI**.

- A **Representational State Transfer (REST) API** is used on the **Controller** as an interface to interact with it.
- Data is sent in a structured (serialized) format such as **JSON** or **XML**.
 - This allows apps and programs to read and use the data.



Visualization of the **Northbound Interface (NBI)** being used by a program in order to receive information from the **Controller**

Automation in Traditional Networks VS Software-Defined Networking (SDN)

Traditional Networks

Networking tasks can be automated in traditional networks, for example:

- Scripts can be written (using Python for example) to push commands to many devices at once.
- Python with good use of **Regular Expressions** can parse through **show** commands to gather information about the network devices.

Software-Defined Networking (SDN)

The Controller collects information about all devices in the network.

- **Northbound APIs** allow apps to access information in a format that is easy for programs and apps to understand. (**JSON**, **XML**, etc.)
- The centralized data facilitates network-wide analytics.
- **SDN** tools can provide the benefits of automation without the requirement of third-party apps and scripts.
 - You also don't need expertise in automation to make use of **SDN** tools.
 - However, **APIs** allow third-party applications to interact with the **Controller**.